

VADIM SMIRNOV

Machine Learning | NLP | Data Engineering

PROFILE

Middle Data Engineer with experience in ML data platforms and speech synthesis. Research interests: hallucination detection, uncertainty estimation and interpretability in reasoning language models.

EXPERIENCE

Yandex — Self-Driving Group

Feb 2025 – Present

Middle Data Engineer

- Built classifiers using engineered features and embeddings to score autonomous-driving scenes and identify low-value data for storage optimization.
- Designed a new storage architecture and data format for **Bag2YT**, reducing RPS by **23%** and disk usage by **13%**.
- Co-developed an ML data repository with deduplication, dataset versioning and unused-data tracking; streamlined dataset management for the ML team and saved **5 PiB** of disk space.
- Optimized the legacy library for working with YT dynamic tables.

T-Bank (Tinkoff)

2025

Machine Learning Intern, Speech Synthesis

- Fine-tuned text-to-speech models with SFT and DPO, increasing crowdsourced quality scores by **40%**.
- Adapted a speech synthesis model to English, improving English-language quality scores by **20%**.

Huawei Technologies

2024

Student Practice

- Built an SVM-based network-optimization prototype and a retrieval-augmented generation (RAG) pipeline.

EDUCATION

Saint Petersburg State University

2023 – 2027

B.Sc. in Data Science, Faculty of Mathematics and Computer Science; fourth-year student.

Yandex School of Data Analysis

2025 – Present

Data Science track.

RELEVANT COURSEWORK AND PROJECTS

- ML and NLP:** classical machine learning, deep learning, embeddings, Transformers, language modeling, fine-tuning, model evaluation and neural network interpretability.
- Mathematics:** linear algebra, calculus, optimization, probability, mathematical statistics, stochastic processes, graphical models and Bayesian inference.

Student projects:

- CycleGAN for Monet-to-photo and horse-to-zebra translation.
- arXiv paper classification with SciBERT.
- Vision Transformer from scratch; Stanford Cars classification with ConvNeXt and EfficientNetV2.

HONORS

- SPbSU Olympiad in Mathematics:** First prize.
- All-Russian Mathematics Olympiad:** Participant, Moscow team.
- Moscow Mathematical Olympiad:** Prize winner.
- Open Mathematics Olympiad:** First prize.
- HSE Vysshaya Proba:** First prize.
- All-Siberian Open Olympiad:** Absolute winner.

TECHNICAL SKILLS

Machine Learning: Python, PyTorch, Hugging Face Transformers, scikit-learn, CatBoost, LightGBM, PyMC.

Data and Tools: NumPy, Pandas, SQL/YQL, YT, Git, Linux, C/C++.

Languages: Russian (native), English (advanced).